

Optical LAB 2: Photometry of Star Clusters

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1. INTRODUCTION

Observing star clusters can teach us a lot about the formation of the Milky Way. We can learn the evolution and age of these clusters, which has a large range. The Milky Way contains almost 200 globular clusters, which tend to be very old, so they are ideal for studying the early formation of our galaxy.

2. OBSERVATIONS

I have decided to analyze data taken on Messier 2 (also referred to as M2 or NGC 7089) which is a globular cluster. This cluster was the one that my group (Adam, Mihir, Sapphire, and myself) observed, so I wanted to work with the data that I had hands-on experience collecting. After all of the struggles and hard work to get these data, it would feel distasteful to abandon them. Before arriving to the telescope, our group had a list of preferred clusters to observe, including M13, M71, and M92. Upon working with the telescope, we noticed that some of these were not attainable due to the dome setup, so we landed on M2.

We began by collecting bias frames (20 with 0.00001 second exposure times) and flat-field images (5 per each of g, r, and i filters with 5s, 1.5s, and 1s exposure times, respectively) for later calibration. Then, we began collecting data on M2. This was a process of trial and error which resulted in a total of 46 frames of M2. Within each filter (g, r, and i), we experimented with different exposure times (10, 15, 30, 45, and 60 seconds). We found that longer exposure times led to some trailing of the stars (we attributed this to a tracking issue with the telescope) and some results appeared blurry. From these 46 files, I was able to narrow them down to the best ones for 15s and 30s of each filter. For the purposes of analyzing the data, I will work with the 30s exposure time taken in each of the g, r, and i filters.

3. DATA ANALYSIS

Note: The Python code used for analysis is shown at the end in the Appendix (Section 5).

3.1. Source Identification and Photometry

As we discussed in Optical Lab 1, we must reduce our science images with the biases and flats to account for noise from the CCD. First, I calculated the median of all 20 bias frames to create a master bias (shown below in Figure 1).

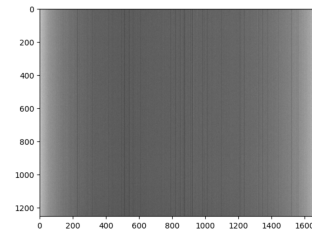


Figure 1. Master bias frame

Next, I created a normalized flat-field image of average intensity equal to 1 for each filter. To accomplish this, I separated the flats by filter (g, r, and i) and took the average of the five flats of each filter, resulting in three master flats. I then subtracted the master bias frame from each master flat. Lastly, I divided each bias-subtracted master flat by the average of this bias-subtracted master flat to create the final normalized flat-field images. For each one, I calculated the mean of the normalized flat to ensure that the average intensity was equal to 1 (and they all were). To finally reduce the science images we collected, I took each image and subtracted the master bias, then divided by the normalized flat for the corresponding filter. The results of the reduced science images to be used for the rest of analysis are displayed below (Figure 2).

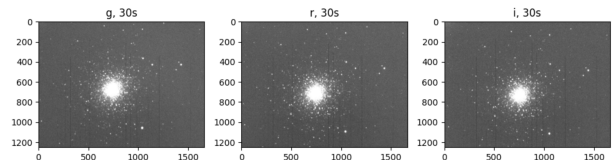


Figure 2. M2 in g, r, and i filters

To subtract the sky background from our science images, I first had to estimate a value for the sky back-

ground. I used sigma clipping (removing outliers that are a specified number of standard deviations away from the center value) on each image. I set a limit of ± 3 standard deviations such that all pixels above or below 3σ from the image's median value will be removed. The calculated median following this clipping will be the estimate of the sky background. These values are shown below, by filter, in Table 1.

Filter	Estimate
g	795.6561
r	679.0807
i	201.3992

Table 1. Sky background estimates

For each science image, I extracted photometry for all sources of interest; in this case, sources that are 5 times greater than the image noise. Using the DAOSStarFinder function in Python, I was able to generate a list of sources for each sky-subtracted image. The DAOFIND algorithm works by searching these images for local density maxima having peak amplitude greater than a specified threshold (as mentioned, I set this to be 5 times above the noise of each science image). It also compares Gaussian kernels, defined by a few parameters of which I specified only the full-width half-maximum (FWHM) to be equal to 3. This source detection procedure returned a table for each science image containing a list of sources and information including x- and y-centroid, flux, and magnitude. This method found 198, 364, and 354 sources for the g, r, and i filters, respectively. From these tables, I created scatterplots of the source positions based on the x- and y-centroids. These are shown in Figure 3, plotted with $\alpha = 0.5$ to see any overlapping sources.

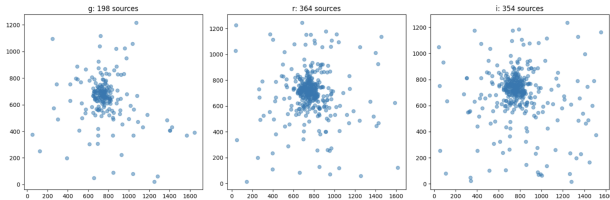


Figure 3. Source detection in g, r, and i filters

I ran aperture photometry to calculate the flux from each source, summed over the entire aperture. Based on the positions of each source (x- and y-centroids as discussed above), I created a circular aperture with a radius of 4 pixels. Applying aperture photometry to each sky-subtracted science image yielded a table with

information about the summed aperture for each source. From these values, I calculated the magnitude of each source such that

$$m = -2.5 \log_{10}(I_*)$$

where m is the calculated magnitude and I_* is the intensity (summed aperture) of the source. The sky-subtracted science images are shown below in Figure 4 with the calculated circular apertures overplotted.

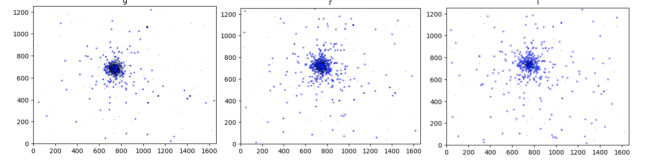


Figure 4. Images with apertures overplotted

I calculated uncertainties for the measured fluxes following

$$N = \sqrt{S_* + n_{pix}(S_{sky} + \sigma_R^2)}$$

where S_* is the flux of the source, S_{sky} is the sky background, n_{pix} is the size of the aperture in pixels, and σ_R is the read noise. As calculated in the previous lab, our CCD has a gain equal to 1, so we can use the flux and sky background values directly. To calculate the read noise, I took the standard deviation of the master bias frame. The aperture size would be the area of a circle (as I used circular apertures) with a radius of 4, so $\pi 4^2$. The flux of the source is the aperture sum and the sky background is as is calculated above. I calculated noise for each filter and plotted the signal by noise for each (Figure 5).

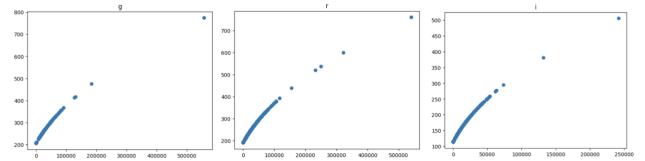


Figure 5. Signal to noise

It appears that the g filter has lower signal-to-noise, but there were also considerably less sources found when using this filter (198 compared to 364, 354). Other factors contributing to uncertainty and poor data quality could be the exposure times used.

3.2. Color-Magnitude Diagram

I begin here by catalog matching sources in two filters: r and i. As above, these two filters returned a relatively

close number of sources and found more than the g filter. First, the positions of the sources had to be converted to celestial coordinates (RA, DEC). Then, the sources found in the r band are matched to the closest one found in the i band. Then, I created a table sorted by the matched sources in the r and i bands. The total number of matched sources was 364.

From these matched sources, I can plot a color-magnitude diagram where color is the magnitude difference between the r and i filters. The plot shows brightness increasing as the y-axis moves from -4 to -14 and shows temperature decreasing (red color increasing) as the x-axis moves from -10 to +6. The CMD is shown in Figure 6.

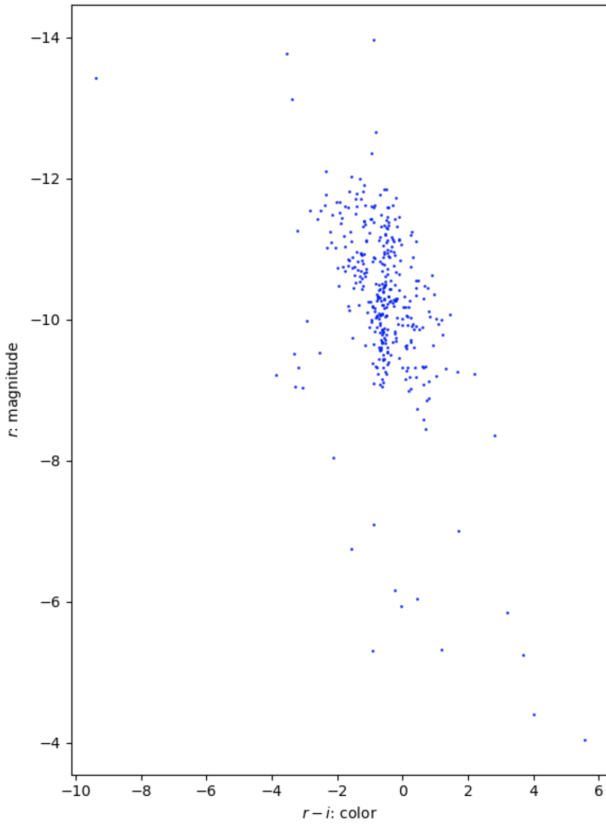


Figure 6. Color-magnitude diagram

I converted flux uncertainties to magnitude uncertainties via

$$\sigma_m = 1.086\left(\frac{\sigma_I}{I}\right)$$

where I =signal, σ_I =noise for a given source.

4. CONCLUSIONS

The main outcome from this analysis was achieving the (almost) end result of plotting a color-magnitude diagram. Although I calculated the uncertainties in mag-

nitude, I was not able to figure out how to incorporate these into the diagram.

I was able to find another research team's CMD of M2 using the r and i bands. They plotted the i band magnitude on the y-axis, so I replotted mine for better comparison (Figure 7).

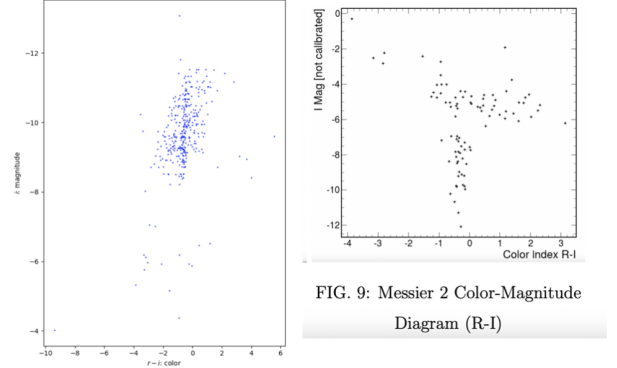


Figure 7. Color-magnitude diagram: Comparison

The overall shape of the sources seems to be similar, with a vertical range of stars and some trailing off horizontally to the right.

In the future, I would like to spend more time overplotting with isochrones and trying to estimate the age of this globular cluster.

While the data collection process for this lab was the trickiest one yet, it was inspiring to have the ability to observe these clusters.

5. APPENDIX

A. Python code is attached.